

Thomas Ratsakatika

Programme leader and machine learning researcher | economic growth and nature

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My work sits between research, engineering and policy. I spent ten years with the UK Government, including postings to British embassies and high commissions in DR Congo, Sierra Leone and Tanzania. I led teams of technical specialists and programme managers and secured over £85m for infrastructure, climate and job-creation programmes. I returned to Cambridge in 2023 to specialise in machine learning and remote sensing, leading research on old-growth forest detection and deploying an AI-based wildlife monitoring system that has been in operational use since 2024.

Experience

University of Cambridge

Sep 2023 – Present

PhD and MRes Researcher, UKRI Centre for Doctoral Training in AI for Environmental Risks (AI4ER)

- **Lead author of a study** testing whether geospatial foundation-model embeddings (AlphaEarth, TESSERA) outperform conventional Sentinel-1/2 features for detecting old-growth forest across 211,893 ha of Romania's Făgăraş Mountains, using spatially buffered nested cross-validation with XGBoost and CNNs. Open data and code.
- **Built and deployed a real-time wildlife alert system** with Fundația Conservation Carpathia. Photos from 4G camera traps pass through a two-stage pipeline (MegaDetector detection, DeepFaune species classification) that alerts rapid intervention teams. In operational use since May 2024.
- **Teaching:** supervisions for second-year geography undergraduates on Global Environmental Change: Ecosystems, Carbon, Climate and Remote Sensing, and dissertation workshops on quantitative methods and remote sensing.

Biodiversity Challenge Funds

Jun 2024 – Present

Expert Committee Member

- **Strategic and funding advice** for the Darwin Initiative, the UK's grant scheme for biodiversity and local communities.

Foreign, Commonwealth & Development Office

Sep 2013 – Jul 2023

Head of the Private Sector Development Hub, British High Commission, Dar es Salaam, Tanzania

Aug 2020 – Jul 2023

- **Led the UK's partnership with Tanzania on climate finance and private sector development**, managing a team of technical specialists and programme managers.
- **Led diplomatic engagement ahead of COP26**, after which Tanzania doubled its mitigation target, removed coal from its energy master plan and committed to two major forest declarations.
- **Secured ministerial approval for a £35m green growth and job-creation programme** that uses grant incentives to leverage larger investments and supports policy and regulatory reform.
- **Ran a pilot using AI on satellite imagery to detect mangrove loss** in the Rufiji Delta, with Tanzania's National Carbon Monitoring Centre and Omdena.

Deputy Team Leader, Economic Growth, British High Commission, Freetown, Sierra Leone

Jul 2017 – Aug 2020

- **Pivoted the UK's post-Ebola economic partnership** from short-term recovery towards long-term growth and jobs.
- **Secured ministerial approval for Invest Salone**, a £30m programme supporting agribusiness, tourism and fisheries, and led its design, procurement and mobilisation.
- **Oversaw a US\$50m public-private partnership to install and operate 100 solar mini-grids**, one of the largest of its kind in Africa at the time; advised on the off-grid regulatory framework.
- **Deputised for the Team Leader**; line-managed, recruited and mentored team members.

Private Sector Development Adviser, British Embassy, Kinshasa, DR Congo

Aug 2015 – Jul 2017

- **Managed a £70m portfolio** of agriculture, renewable energy and public works programmes in a conflict setting.
- **Led a £50m market systems programme** in agriculture, river transport, renewable energy and financial services, reaching an estimated 979,000 people.
- **Led the redesign of a struggling £20m road programme** in the conflict-affected east; construction provided temporary employment for 2,000 people.

Private Sector Development Adviser, Investment Climate Team, Glasgow, UK

Sep 2014 – Aug 2015

- **Managed two research programmes**, delivered by the World Bank and an India-based NGO, building evidence for competition policy reform in Africa and Asia; represented the UK at conferences in Bangkok and Nairobi.

Economic Analyst, Chief Economist's Office, London, UK

Sep 2013 – Aug 2014

- **Modelled global poverty trends, economic complexity and poverty-adjusted discount rates** in a team led by Professor Stefan Dercon; organised a seminar series with leading academics.

Aon Investment Consulting

Sep 2010 – Aug 2012

Associate Consultant, London, UK

- **Built financial models across the major asset classes**; contributed to research on global market trends, including emerging market debt and the European debt crisis; administered a £240m derivative fund.

AgustaWestland (now Leonardo)

Jul 2007 – Jun 2008

Aerospace Engineer (industrial placement), Yeovil, UK

- **Modelled a vibration reduction system in MATLAB/Simulink** and cracking in composite rotor blades; received a company award for automating reporting processes resulting in an 18-fold efficiency gain.

Education

Exp. 2027	PhD Machine Learning and Remote Sensing , University of Cambridge Thesis on old-growth forest detection using machine learning and remote sensing data. UKRI doctoral studentship in AI for environmental risks. Co-supervised across the Geography and Computer Science and Technology departments.	
2024	MRes Environmental Data Science , University of Cambridge Machine learning, statistics, environmental data analysis. Automated wildlife monitoring research.	
2021	CFA charter awarded, CFA Institute (CFA Program completed 2012)	
2013	MPhil Engineering for Sustainable Development , University of Cambridge	Distinction
2010	MEng Aerospace Engineering , University of Bath	First Class

Publications and talks

2026	Ratsakatika, T. , Zotta, M., Keshav, S. and Lines, E.R. Geospatial embeddings detect old-growth forests but buffered spatial validation narrows their advantage over Sentinel features. arXiv:2609.28194 (under review). doi.org/10.48550/arXiv.2609.28194	
2026	Ratsakatika, T. , Zotta, M., Keshav, S. and Lines, E.R. Old-growth forest reference labels and model predictions for the Făgăraș Mountains, Romania (2020, 10 m), version 1.0.0 [dataset]. Zenodo. doi.org/10.5281/zenodo.22693148	
Sep 2026	Old-growth forest detection with geospatial foundation models and spatial validation. Talk, Joint NSO–GfÖ Conference, Odense, Denmark.	
Jun 2026	Detecting old-growth forests with geospatial foundation model embeddings and spatial validation. Talk, ML4EO: Machine Learning for Earth Observation, Exeter, UK.	
Jul 2024	Real-time wildlife monitoring with AI and 4G camera traps. Talk, Energy and Environment Group, Department of Computer Science and Technology, Cambridge, UK.	

Skills

- **Leadership and management:** accountable for portfolios of up to £70m; advising ministers and senior officials; team leadership, line management, recruitment and mentoring; contract negotiation and risk management in fragile and conflict-affected settings.
- **Policy and delivery:** diplomatic engagement and influencing, programme design and business cases, public-private partnerships.
- **Research software:** agentic coding; reproducible pipelines with Git, pytest and pre-commit; open code releases.
- **Machine learning:** Python, PyTorch, scikit-learn, XGBoost, CNN, ViT, Optuna; spatial cross-validation and spatial statistics; computer vision (e.g. MegaDetector, DeepFaune, SpeciesNet, Segment Anything).
- **Remote sensing:** Sentinel-1/2, Planet Labs, Google Earth Engine, geospatial foundation-model embeddings (AlphaEarth, TESSERA), rasterio, geopandas, xarray.